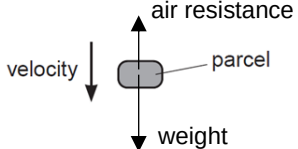


1a	Taking right as positive, $s = ut$ $4.40 = 5.40 \times t$ $t = 0.8148 \text{ s}$ Taking downwards as positive, $s = ut + \frac{1}{2}at^2$ $h = \frac{1}{2} \times 9.81 \times 0.8148^2$ $= 3.2565 \approx 3.26 \text{ m}$	C1 A1
1bi	 downward pointing arrow labelled weight upward pointing arrow labelled air resistance no credit if magnitude of air resistance exceeds that of weight	B1
1bii	Air resistance increases (as velocity increases) Weight (or mass) is constant, so resultant force decreases Hence, acceleration decreases	B1 B1 B1
1biii	At terminal velocity, air resistance equals weight With larger mass, weight is larger. (Greater air resistance), so greater terminal velocity	B1 B1
1biv 1.	gravitational potential energy to kinetic energy and thermal/internal energy	B1
1biv 2.	gravitational potential energy to thermal/internal energy	B1